

Engineering Curves: Construction of Hypocycloid

Rolling Circle on Interior of Fixed Circle with Tangent & Normal

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Problem Statement

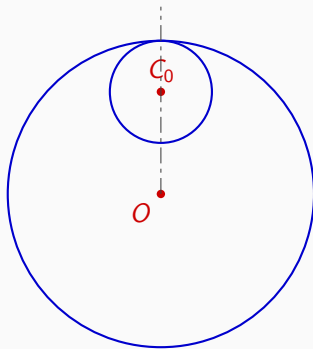
Question

Draw a **hypocycloid** generated by a rolling circle of diameter $d = 30$ mm rolling on the inside of a fixed base circle of diameter $D = 90$ mm for one complete revolution. Construct a **tangent** and a **normal** at any point P on the curve.

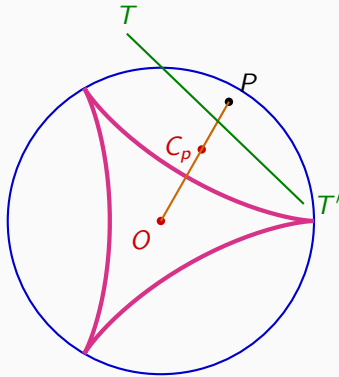
Step 1: Draw Fixed Base Circle & Internal Rolling Circle

Base Circle → Radius $R = 45$

Rolling Circle → Radius $r = 15$



Step 2: Complete Hypocycloid Curve with Tangent & Normal



Hypocycloid → Generated Curve

$N-N'$ → Normal Line

$T-T'$ → Tangent Line